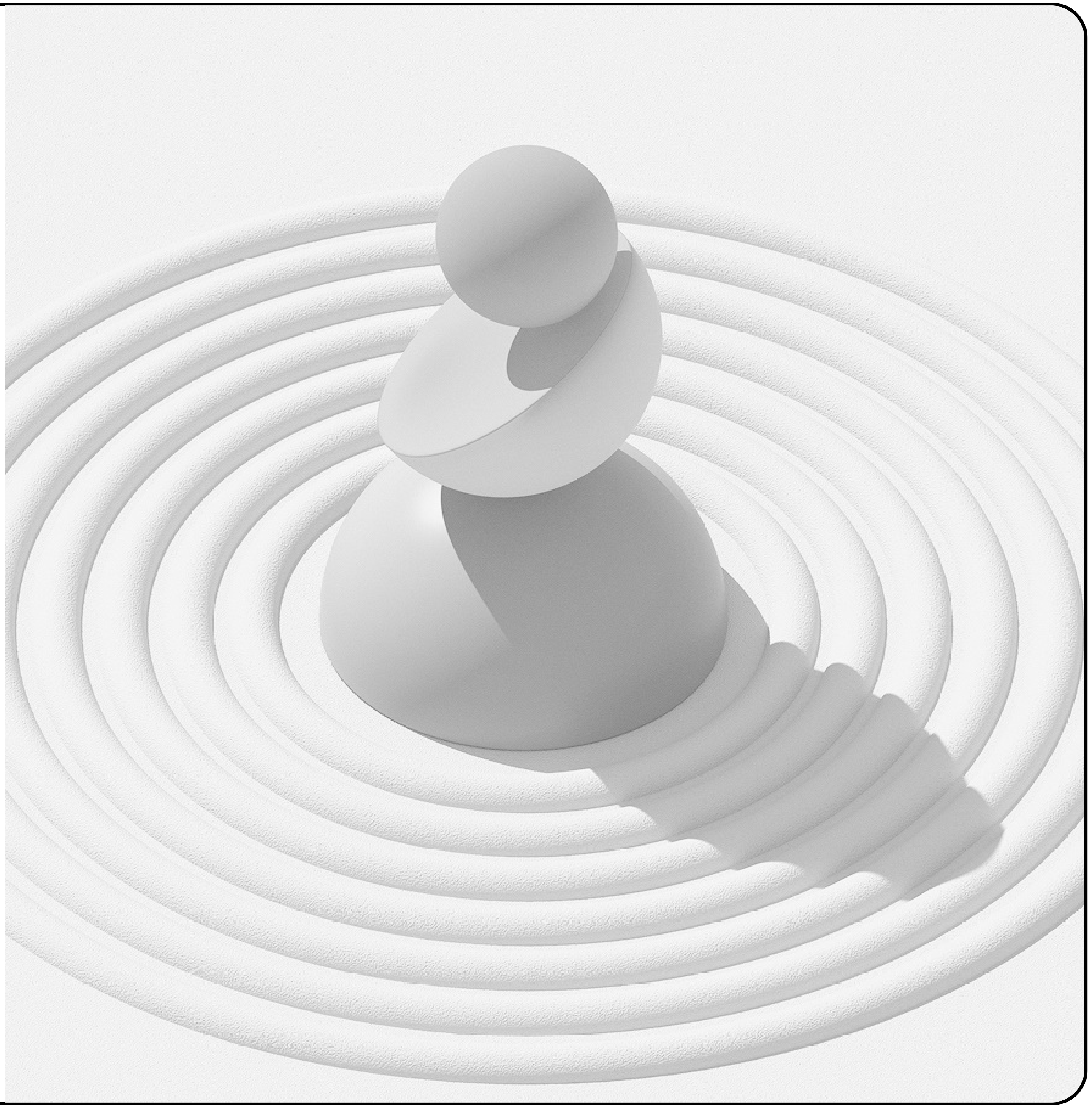
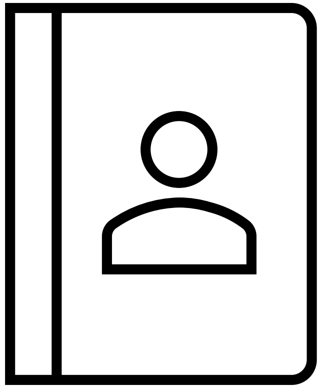
ECON251 INDUSTRY AND TRADE IN ASIA

Case study Sample B



See my key observations for each chart

KEY GUIDELINES



R2 (0.5) : reliability of trendline / report suggesting how many % of the points are explained in the line/ report ;

P (< 0.05) : > 95% confidence level of this variable plays the stated role in the regression

1. Spend time run through each chart even before you read the questions

2. Pay close attention to
   a. The units of measurement in each axis esp those with %
   b. The secondary axis
   c. Values below 0 axis
   d. The trendline and details of the equation [not all trendlines include the equation]

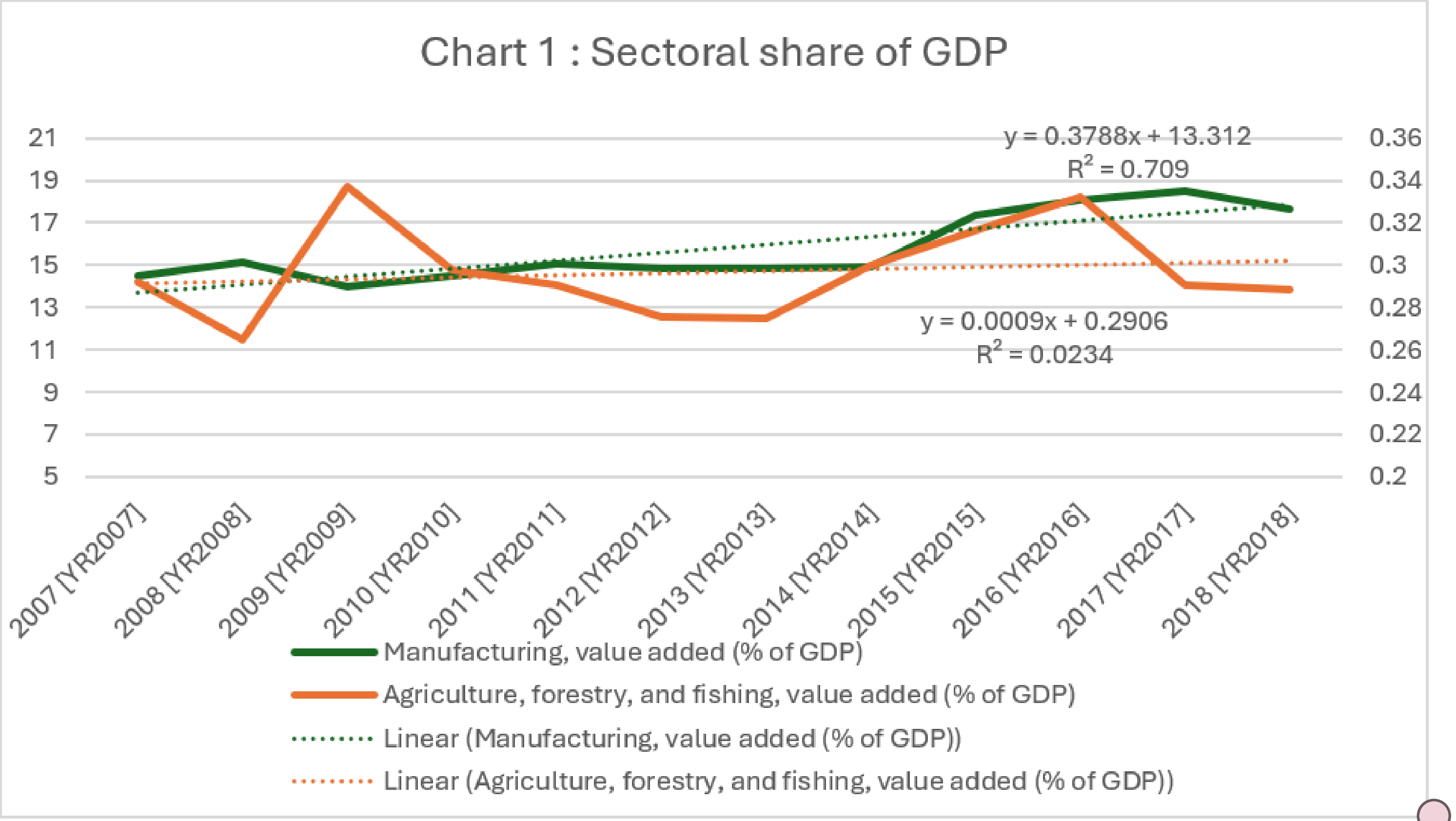
3. You are not required to state the precise values from the chart

4. Each question usually carries 4- 6 marks so ensure you answer ALL the questions.

5. When answering the case study questions, please note that the questions follow the chronological order of the charts provided in the appendix.

   - **Sequential Order:** You will extract evidence progressively (e.g., if a question requires evidence from Chart 1, the subsequent question will draw from Chart 2, and so on).

6. You are not expected to have prior knowledge of the Asia economy. Focus is still on the key concepts covered in lectures on the general approach of growth of the Asia economies



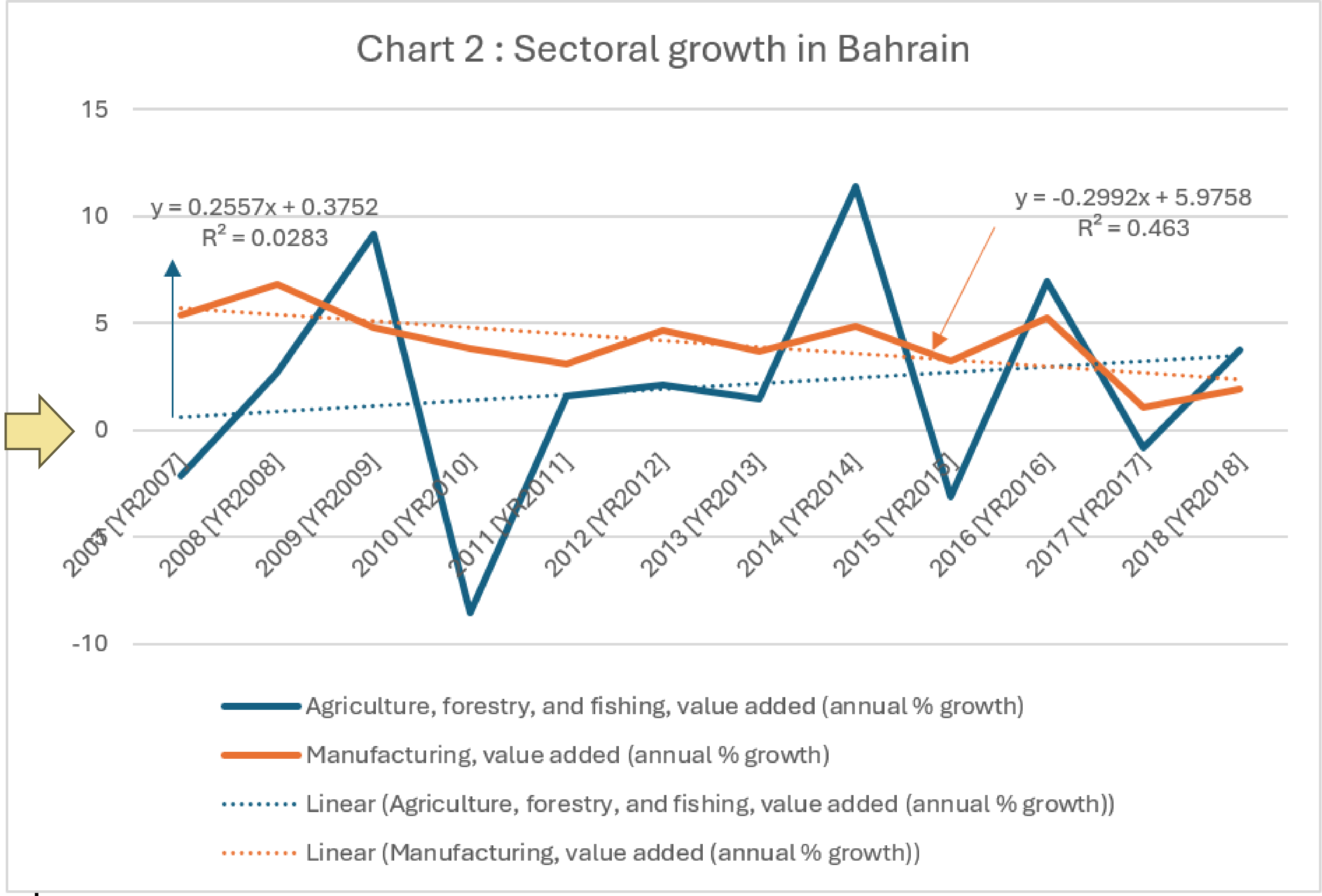
You scribble short notes of such in your qn paper

Evidence of Industrialisation

Ie firms moving away from agriculture and into industry / manufacturing sector

Key observations

a. Axis: % share of GDP : how much they contribute to country's output
b. Share of Manu and Agri trendline both upward sloping but only trendline for Manu with R2> 0.5 is reliable
c. Share of Manu much larger (double digit) than share of Agri (< 1 %)
d. data of Manu thus indicate the economy is advancing in industrialisation



Key observations

Growth rate of Manu all above 0% , i.e. output in this sector is growing despite it shows a unreliable falling trend (R2 < 0.5)

Growth rate of Agri rising but R2 < 0.5 so not reliable. And also there are occurrence of <0 growth rate suggesting Agri output falls

Answers to Question 1

What are the key evidence we can gather that there is signs of industrialisation in Bahrain [4 marks]

Answers :

Industrialisation suggest an economy is moving its production towards the industry sector (esp. manufacturing) and away from the agricultural sector

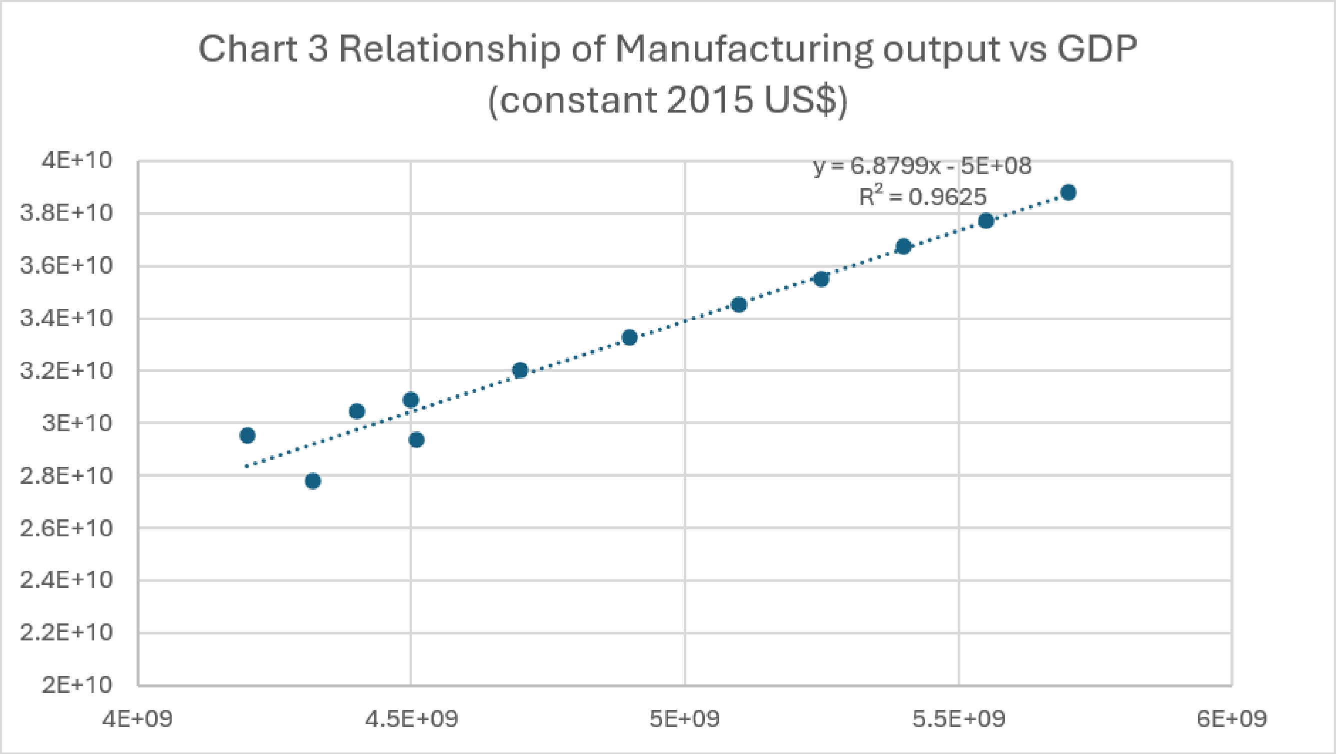
Data in chart 1 shows manufacturing sector contribute **more** to Bahrain's GDP and the sector's share of GDP is also rising with a reliable trendline [1 mark]

Data in chart 2 shows the output growth of Manu is rising throughout the period given the growth rates are all >0. Albeit a falling trendline but it is unreliable and also suggest output by this sector is still growing [1 mark]

Data in chart 1 and 2 of agriculture shows its more volatile (as seen in chart 2) and a smaller contribution to the GDP [1 mark] but this indicator is not falling as given in the traditional way .

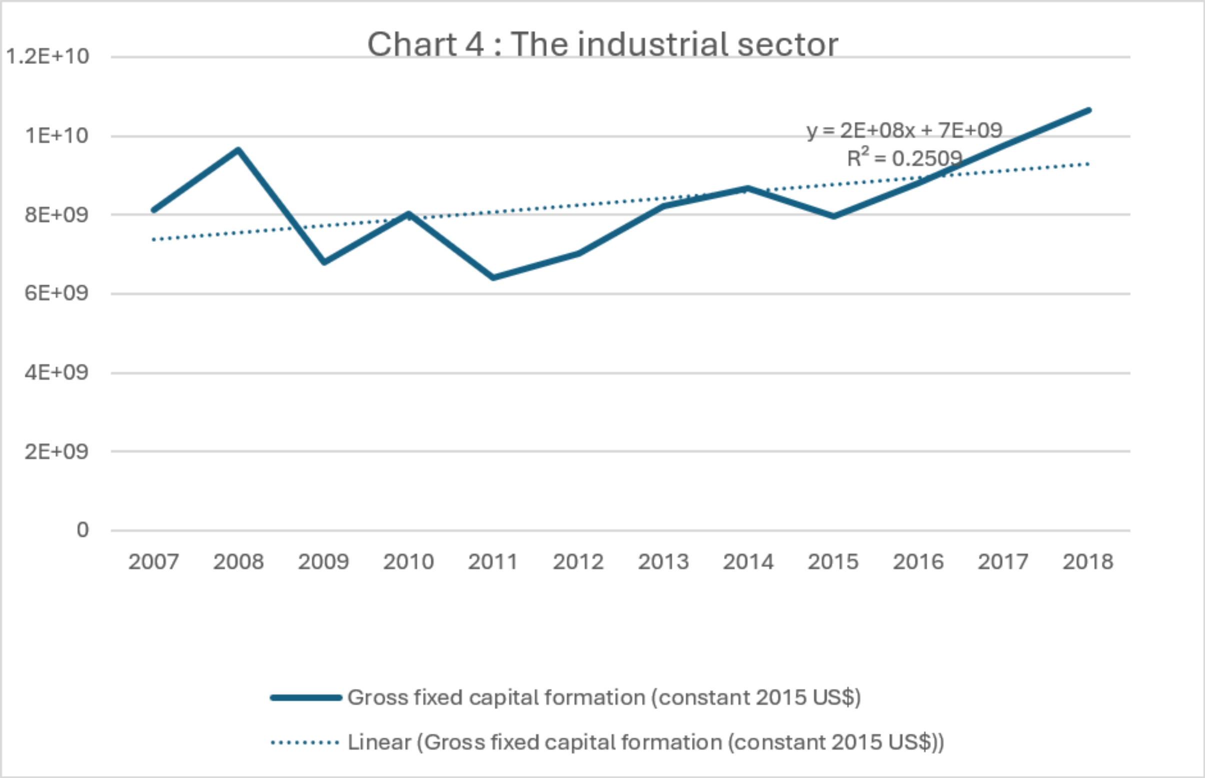
Thus it is **conclusive that Bahrain is experiencing industrialisation** in the studied period based on the manufacturing data [1 mark]

Write in short sentences referring to the evidence and concepts
Use multiple paragraphs



Key observations

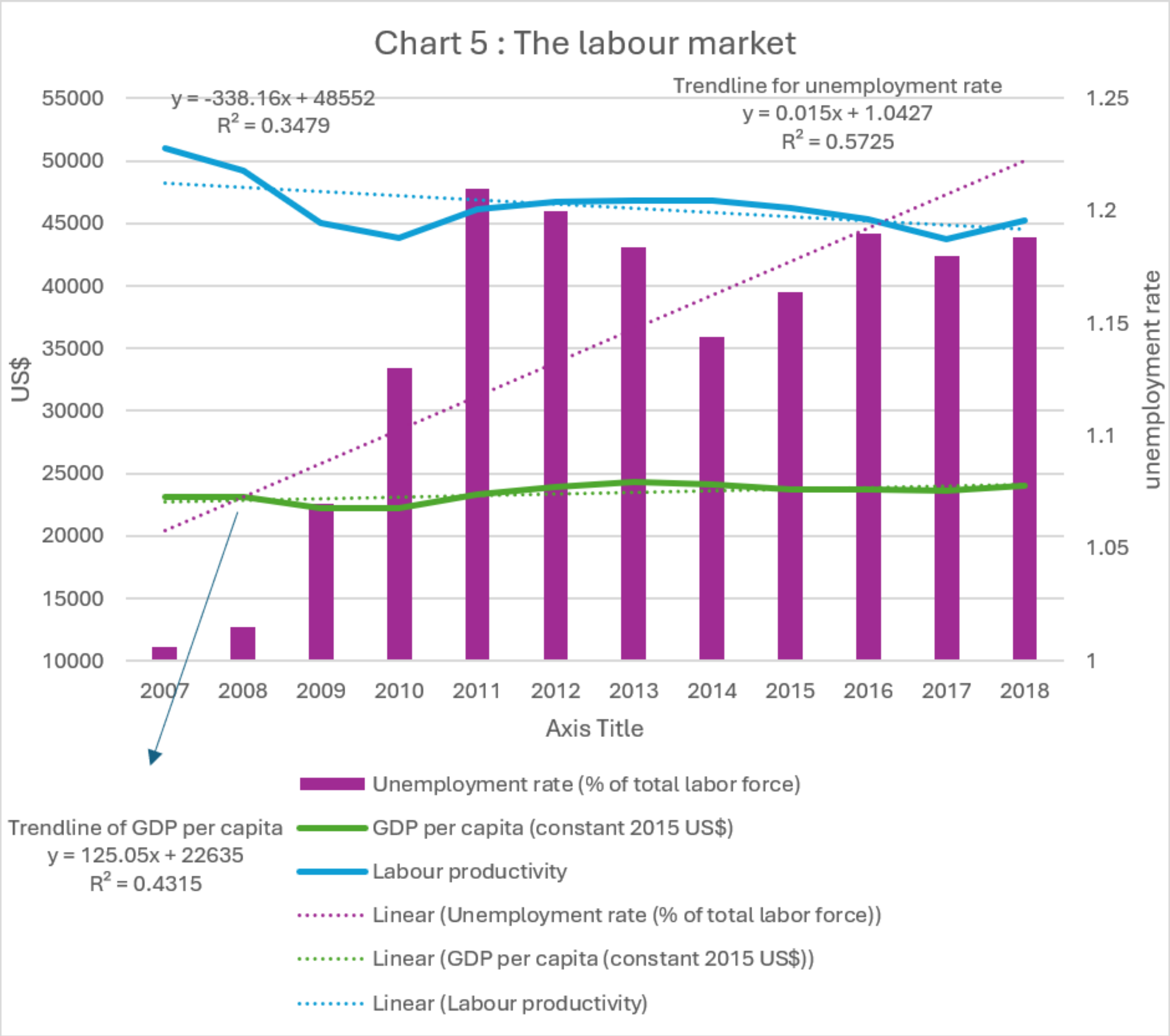
The scatterplot shows a positive relationship between the growth of Manu output on the GDP
This positive relationship is reliable with a very high R2
This suggest industrialisation is beneficial to the economy



Key observations

In the studied period, data here shows a rise in fixed capital formation This is expected as firms acquire more equipment and machines for industrial production

However we cannot draw this as a strong evidence given the low R2 (< 0.5)



Key observations

1. Labour productivity
A falling trend but low R2: suggest labour productivity is not falling most of the time. A weak evidence

2. GDP per capita
Suggest a rising trend but not reliable with R2 <0.5. A weak evidence

c. Unemployment rate
Rising trend with strong reliability but it DOES NOT suggest a benefit to the economy . Rising unemployment means workers cannot obtain jobs due to incompatible skills or lack of jobs

Answers to Question 2

What are the key evidence we can gather that on the key benefits of industrialisation in Bahrain [5 marks]

Answers :

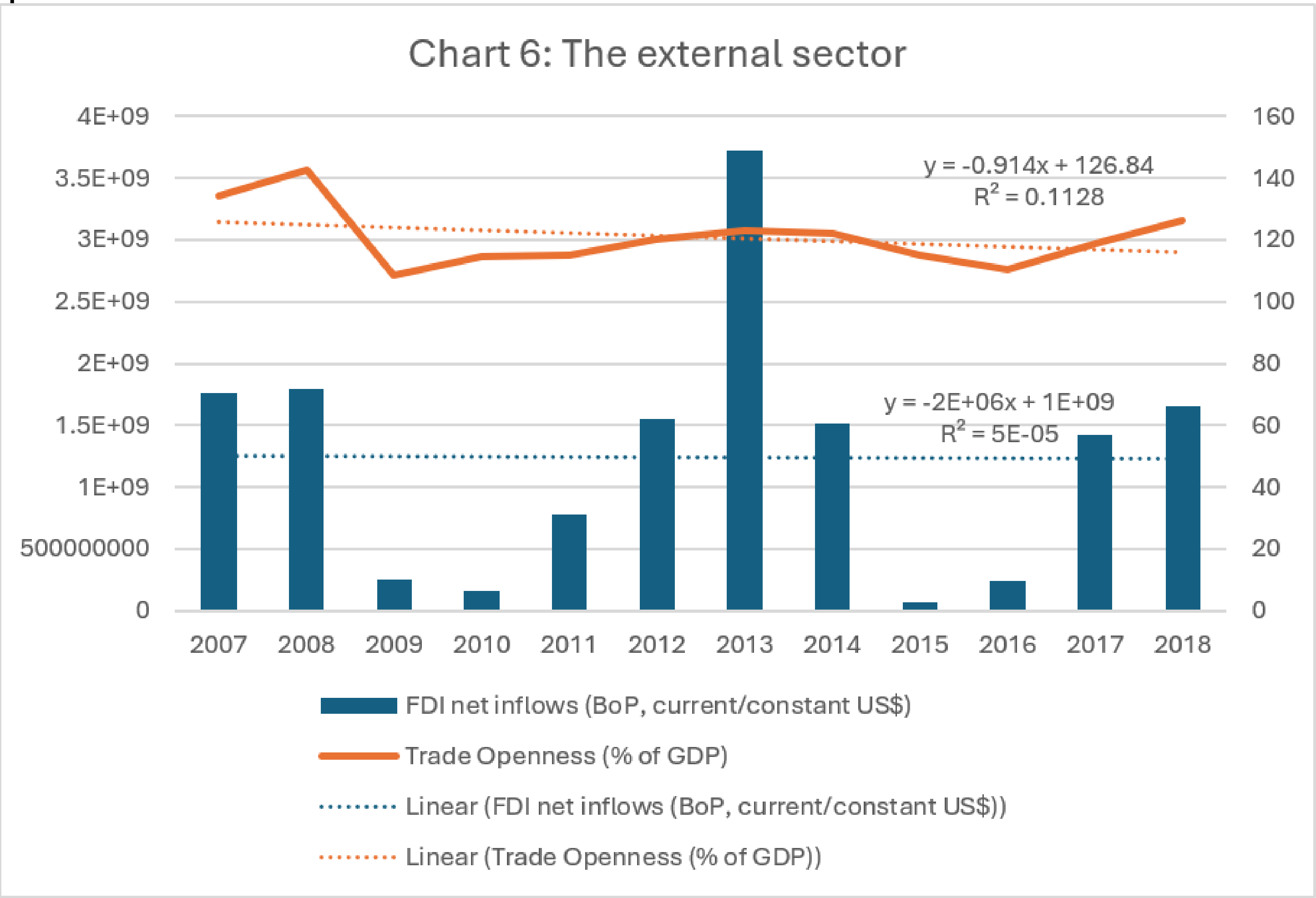
Benefits of industrialisation are in terms of

1. Rising GDP as given in chart 3 'This is a strong evidence given the very high R2 [2 marks]

2. Other benefits but as weak evidence as their R2 are low [3 marks]

a. Rising fixed capital formation
b. GDP per capita
c. Labour productivity

Thus we see that in Bahrain, the benefits of industrialisation are not passed on to the labour market



Key observations

1. Trade openness above 50% suggesting very high volume of export and imports flowing into the economy

Despite a unreliable falling trend it is still a good evidence of an open economy

2. FDI inflow fluctuates as R2 is close to zero 9 i.e. data points are regularly above or below the trend line)

Answers to Question 3

Explain using the data in the following chart why Bahrain is considered an open economy (i.e. adopting outward oriented growth path)

State 2 other indicators that we will need to further justify the country is an open economy [6 marks]

Answers

Data in chart 6 shows Bahrain is open to trade and thus an open economy . Despite a weak evidence of a falling trend, the very high **trade openness of > 50%** shows the large volume of exports and import flowing [2 marks]

The evidence of FDI inflow serves as a weaker evidence moderate level in most periods. Thus funds are still flowing into the economy suggest, a fairly open economy [2 marks]

Other indicators required [2 marks]

a. Falling tariff rate suggest imports are taxed lesser and thus can flow more easily into the economy
b. GVC rate esp. backward participation shows Bahrain is a participant in the global value chain where she process the semi finished goods tor intermediate parts from the other countries

Regression Statistics	
Multiple R	0.840405629
R Square	0.706281621
Adjusted R Square	0.538442547
Standard Error	705595163.9
Observations	12

ANOVA

	df	SS	MS	F	Significance F
Regression	4	8.38023E+18	2.09506E+18	4.208088173	0.047680067
Residual	7	3.48505E+18	4.97865E+17		
Total	11	1.18653E+19			

P < 0.05 suggest a 95 confidence that the variable plays a significant role in affecting y

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	ower 95.0%	Jpper 95.0%
Intercept	7969569245	13589845883	0.586435587	0.575997152	-24165309905	4.01E+10	-2.4E+10	4.01E+10
GDP per capita (constant 2015 US$)	-629287.6888	873824.7928	-0.720153163	0.494768732	-2695554.986	1436980	-2695555	1436980
Gross fixed capital formation (constant 2015 US$)	0.499183178	0.293403606	1.701353249	0.132667643	-0.194606104	1.192972	-0.19461	1.192972
Unemployment rate (% of total labor force)	-3678935994	4004753012	-0.918642419	0.388839675	-13148672087	5.79E+09	-1.3E+10	5.79E+09
Trade openness (% of GDP)	54428420.54	22059392.58	2.46735808	0.042999118	2266245.859	1.07E+08	2266246	1.07E+08

Key observations

Positive factors
Gross fixed capital formation but P > 0.05
Trade openness with P < 0.05 (Strong driver of FDI)

Negative factors
GDP per capita, unemployment rate with P > 0.05
The increase in these variables will REDUCE FDI inflow

Question 4

a) Explain the role of rising unemployment (as seen in chart 5), in affecting the level of FDI inflow in Bahrain and explain if the government should be worried about this development. [4 marks]

Answers :

Chart 5 shows rising unemployment in Bahrain, and this may occur as the economy moves rapidly to the manufacturing sector
This suggest workers are not equipped or skilled to find jobs in the new sector .
This suggest government should be worried about this development

However, the evidence of it reducing FDI is weak as this variable's p value > 0.05, so we cannot fully deduce this rising unemployment will have a strong impact in Bahrain . Perhaps a much longer duration of studies, a more detailed variable [e.g. unemployment in the manufacturing sector] will provide a more robust study.

b) With reference to Chart 6 and the regression results, Bahrain's FDI inflows are highly volatile. Identify the most important positive x variable driving this relationship, and explain one economic reason for its impact. [7 marks]

Answers

The most important positive x variable that will drive FDI inflow into Bahrain is trade openness [1 mark]

The regression results suggest a rise in trade openness by 1 % will bring in about $54m of FDI into the economy [2 marks]

The economy with trade openness of > 50 % (as seen in chart 6) suggest an ease of flow of imports and exports. This suggest the foreign companies face lesser difficulties if bringing in foreign parts and inputs for the production. At the same time, export volume may be high suggesting Bahrain a reputable and reliable export partner. This will thus lure foreign firms to set up operation in Bahrain [2 marks]

But a high trade openness may also suggest she is more prone to global crisis , affecting supply chain, and/or global demand, thus making its FDI flow volatile as given in chart 6. [2 marks]